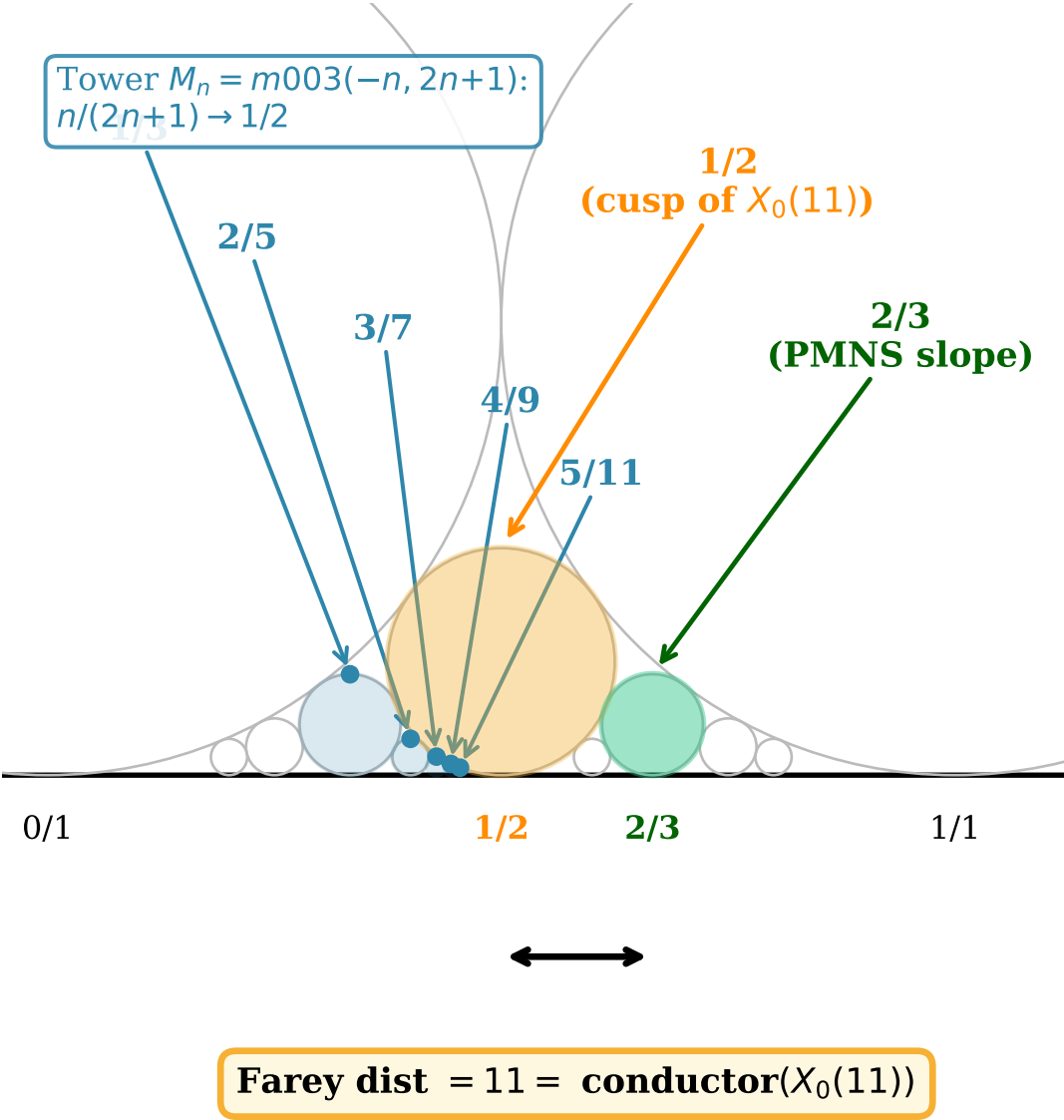


Farey graph: HFG slopes at Farey distance 11



Hecke eigenvalue verification: $X_0(11)$ base change

Norm	Type	a_p $X_0(11)$	Formula	Bianchi K_P	Hilbert K_C	
2 (x2)	split both	-2	-2	-2	-2	OK
9	inert K_C	-1	1-6=-5	--	-5	OK
13 (x2)	split both	4	4	4	4	OK
17	ram. K_C	-2	-2	-2	-2	OK
19 (x2)	split both	0	0	2	0	--
25	inert BOTH	1	1-10=-9	-9	-9	OK OK
43 (x2)	split both	-6	-6	-4	-6	OK
47 (x2)	split both	8	8	8	8	OK
49	inert K_C	-2	4-14=-10	--	-10	OK
67 (x2)	split both	-7	-7	-7	-7	OK

$K_P = \mathbb{Q}(\sqrt{-3})$ (PMNS), $K_C = \mathbb{Q}(\sqrt{17})$ (CKM).
Yellow row: inert prime 5 gives $\mathbb{Z}/5$ torsion fingerprint $a_{(25)} = -9$.